

QUESTION 01

OPTION 02:- Real-World Data Analysis & Business Problem Solving (Open Domain)

INTRODUCTION

My project focus on analyzing Amazon bestselling books from 2009 to 2019 to understand how pricing strategies influence customer satisfaction and popularity. This data set includes 550 records including factors such as user rating, price, reviews, book name, author and genre. The objective of this study is to identify patterns, perform clustering analysis and develop a predictive model to support business decision-making in the book publishing industry.

DATA COLLECTION

The data was collected by publically available source on Kaggle which contains information about Amazon bestselling books over 10-year period. The data set includes key variables such as book title, user rating, Author, Number of reviews, price, year and genre. Since the data set was already structured and clear, it was directly used for analysis.

DATA CLEANING & PREPROCESSING

The data was examined for missing values and inconsistencies. There were no missing values found in the data set. Column names were standardized by removing spaces to improve usability during analysis. Data types were verified. This preprocessing step ensured the data was clean and ready for analysis.

```
df.info()
```

[16]

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 550 entries, 0 to 549
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Name             550 non-null   object
1   Author           550 non-null   object
2   User_Rating      550 non-null   float64
3   Reviews          550 non-null   int64
4   Price            550 non-null   int64
5   Year             550 non-null   int64
6   Genre            550 non-null   object
dtypes: float64(1), int64(3), object(3)
memory usage: 30.2+ KB
```

```
df.describe()
```

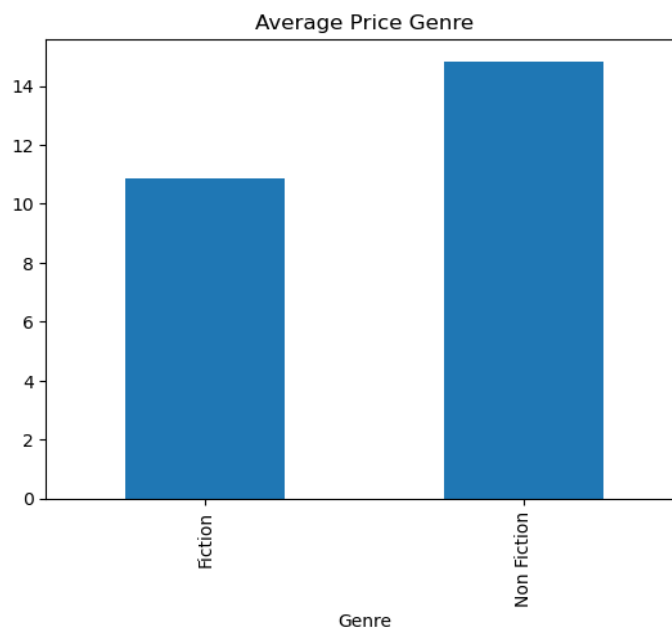
[17]

...

	User_Rating	Reviews	Price	Year
count	550.000000	550.000000	550.000000	550.000000
mean	4.618364	11953.281818	13.100000	2014.000000
std	0.226980	11731.132017	10.842262	3.165156
min	3.300000	37.000000	0.000000	2009.000000
25%	4.500000	4058.000000	7.000000	2011.000000
50%	4.700000	8580.000000	11.000000	2014.000000
75%	4.800000	17253.250000	16.000000	2017.000000
max	4.900000	87841.000000	105.000000	2019.000000

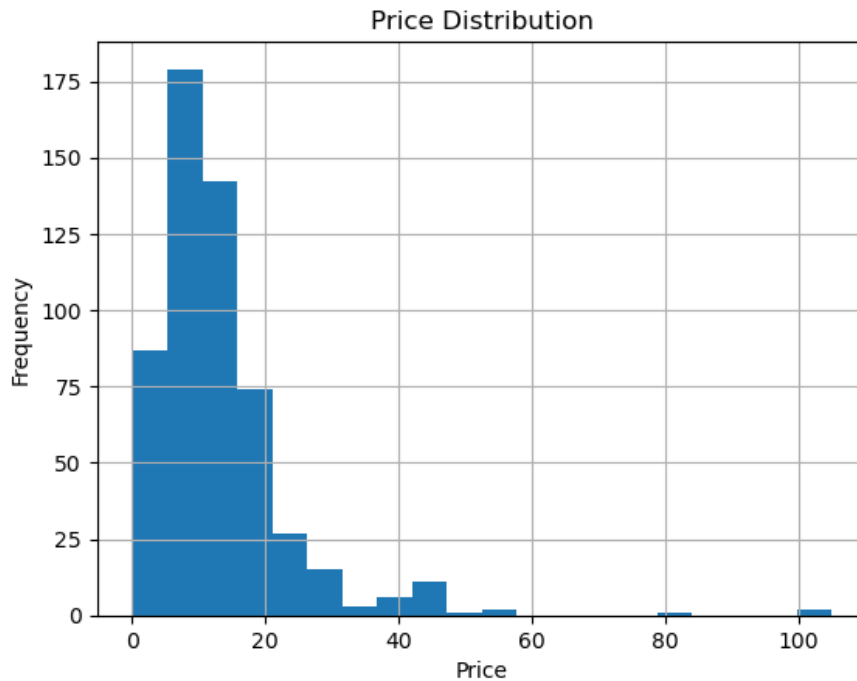
EXPLORATORY DATA ANALYSIS (EDA)

(a) Genre Distribution



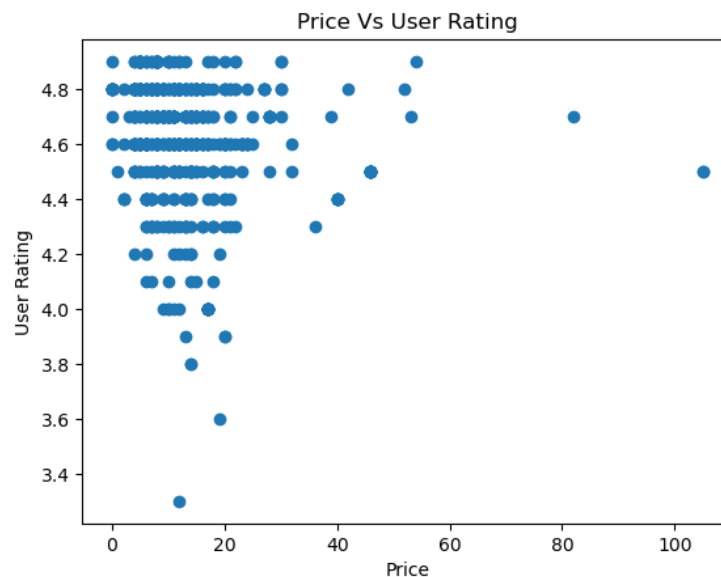
The dataset consists of two categories, which are Fiction and Non Fiction. The distribution shows that one category dominates the bestseller list, indicating reader preference and publishing trends.

(b) Price Distribution



The price distribution shows us that most books are priced within a moderate range. This suggests that publishers follow consistent pricing strategies to appeal a broader audience.

(c) Price Vs User Rating



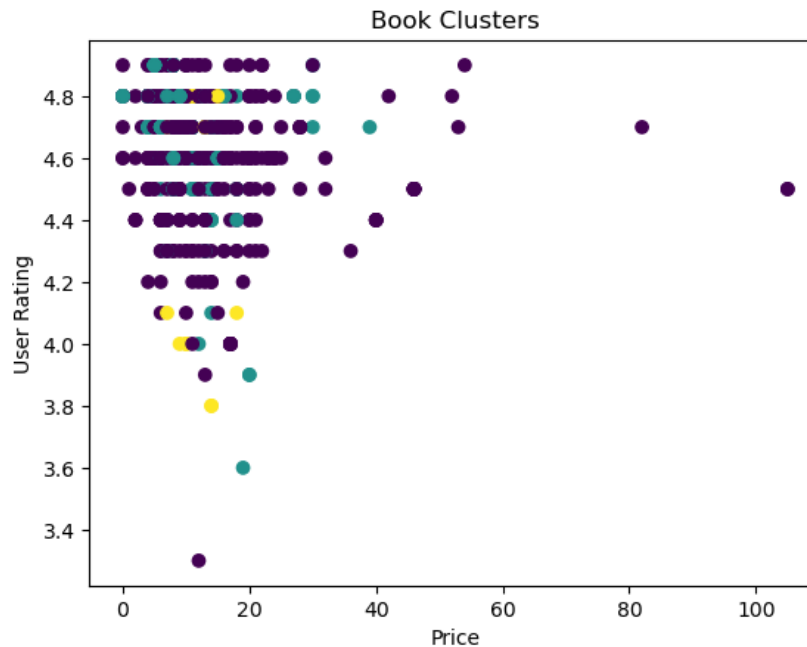
The scatter plot shows no strong relationship between price and user rating. This indicates that higher-priced books do not necessarily result in higher customer satisfaction.

(d) Correlation Analysis



Correlation analysis reveals that price has a weak relationship with user ratings. However, the number of reviews shows stronger relationship with book popularity, indicating that engagement plays a more important role than pricing.

CLUSTER ANALYSIS



K-Means clustering was applied to group books based on price, user ratings and number of reviews. The analysis identified three distinct clusters. One cluster represent highly popular books with high reviews and ratings, another includes moderately perming books and the third cluster consists of books with lower engagement. This part helps understand different types of bestselling books and their market position.

PREDICTIVE MODEL

```
from sklearn.metrics import r2_score, mean_squared_error
import numpy as np

r2=r2_score(y_test,y_pred)
rmse= np.sqrt(mean_squared_error(y_test,y_pred))

print("r2 Scor:" , r2)
print("rmse:", rmse)
```

[45]

```
... r2 Scor: 0.010479440698247466
    rmse: 0.23826597615687772
```

A linear regression model was developed to predict user ratings using price and number of reviews as independent variables. The model was evaluated using R-squared and Mean Squared Error (RMSE). The results indicate that price alone is not a strong predictor of customer satisfaction, while reviews provide better insight into book popularity. The relatively low R-Squared value suggests that additional factors influence user ratings.

BUSINESS INSIGHTS

The analysis highlights that pricing strategies alone do not determine customer satisfaction. Books with higher engagement reflected by the number of reviews tend to perform better regardless of price. Publishers should focus on improving content quality, marketing strategies and customer engagement rather than relying solely on pricing adjustments. Additionally clustering results suggest that different market segment exist allowing publishers to tailor strategies for different types of readers.

CLOUD INTEGRATION (AWS)

In practice, cloud technologies like AWS can be implemented to enhance scalability and effectiveness. Storage of the dataset can be done on AWS S3 for secured and scalable data storage. Processing of the data and training models can be done using AWS EC2 instances, whereas AWS RDS can be utilized for management of structured data. Finally, deployment of machine learning models can be done using AWS Sage Maker. While this project is completed using local technologies, cloud-based architecture provides scalability and real-time analytics.

CONCLUSION

This project successfully analyzed the relationship between pricing, customer satisfaction, and book popularity. The findings indicate that engagement plays a more significant role than pricing in determining success. The clustering and predictive modeling approaches provided valuable insights into market behavior. These results can support data-driven decision-making in the publishing industry and improve business strategies.

QUESTION 02

Probability distribution provides information about the distribution of values in relation to the random variable. In general, probability distributions may be broadly categorized into discrete and continuous depending on the type of values of the random variable under consideration.

First, binomial distribution is a type of discrete probability distribution which measures the number of successes in a certain set number of attempts with the assumption that there are only two outcomes in each attempt – success and failure. The chances of succeeding in any attempt remains constant and the distribution will be used to determine the number of students who passed an examination from a particular set of students.

Secondly, Poisson distribution is a type of discrete probability distribution. It measures the number of occurrences that take place over a particular period of time. The assumptions of this type of distribution include independence of events and consistency in occurrence of the event during the specified period.

Geometric distribution can be described as a distribution for calculating how many experiments are necessary before attaining success in a string of independent experiments. In contrast to binomial distribution, in geometric distribution, there is no pre-determined amount of trails. Among other properties, geometric distribution possesses the memoryless property which indicates that the chance of success in further attempts depends neither on previous nor next results.

The main difference between hypergeometric and binomial distributions is the fact that hypergeometric distribution implies that the process of drawing elements from a finite sample involves sampling without replacement while probability of success changes each time an element is chosen. Hypergeometric distribution is often employed when calculating quality control.

The negative binomial distribution is one more generalization of the geometric distribution. As opposed to geometric distribution, the former implies the number of successful experiments to reach a certain point rather than one success. Thus, the negative binomial distribution offers greater adaptability.

Exponential distribution is a continuous analogue of geometric distribution. In addition, the exponential distribution is utilized in order to find out the time until the next occurrence in a Poisson process.

The normal distribution is an important form of continuous distribution characterized by the symmetrical shape of its graph. The parameters of the normal distribution are its mean and standard deviation. It models natural phenomena such as people's height, weight, or test results. One of the properties of this type of distribution is that the mean value, median, and mode coincide.

Lastly, chi-square distribution is another example of a continuous probability distribution which is widely applied in various forms of statistical hypothesis testing. It has the characteristic of being positively skewed and depending on degrees of freedom.

In conclusion, different types of probability distributions have unique features as well as specific areas of application. Thus, binomial, Poisson, and geometric distributions belong to discrete probability distributions. Meanwhile, normal and exponential distributions can be considered examples of continuous probability distributions.

Distribution	Type	Main Use	Key Feature
Binomial	Discrete	Fixed trials	Success/failure
Poisson	Discrete	Event count	Rare events
Geometric	Discrete	First success	Memoryless
Hypergeometric	Discrete	Sampling without replacement	Changing probability
Negative Binomial	Discrete	r successes	Generalization
Exponential	Continuous	Time between events	Memoryless
Normal	Continuous	Natural data	Symmetrical
Chi-square	Continuous	Statistical tests	Right-skewed

